

Responses to the Review Comments

Comment 1: The study uses only three synthetic radial networks of one topology family with manufactured load/PV shapes; no real feeder, tap changer, capacitor switching, battery, unbalanced laterals, or RTU/PMU noise is tested, so the title's implication for large-scale distribution networks is broader than the evidence supports. The manuscript should state that the results apply to synthetic radial Volt-VAR benchmarking, not large-scale field networks.

Response 1: We agree and have revised accordingly. The title no longer refers to large-scale networks and now reads "Dynamic Optimal Control of Synthetic Radial Distribution Networks Using Multi-Head Attention Graph Neural Networks". The Abstract, the closing paragraph of the Introduction, and the Conclusions now state explicitly that all findings are limited to synthetic radial Volt-VAR benchmarking and do not extend to field networks; the untested field elements (tap changers, capacitor switching, storage, unbalanced laterals, measurement noise) are listed by name.

Comment 2: The four-head daily loss is only 0.000374155 MWh above the teacher over 24 hours while its P95 latency is 45% higher; the paper resolves micro-watt differences below utility relevance yet pays extra inference time. The manuscript should discuss whether a 0.02 percent loss change justifies 45 percent higher latency versus the teacher, instead of presenting head-count scaling as a finding.

Response 2: We added an explicit relevance check at the end of Section 4.3. The loss gap to the teacher is 0.023% of the teacher's daily loss and the advantage over graph convolution is 0.053%, both below any utility relevance, while the four-head policy pays 45% higher P95 latency than the teacher (and 5.4 times graph convolution's). The revised text states that head-count scaling is not presented as a finding, that the remaining differences are operationally negligible, and that whether they justify the extra inference time is a deployment question rather than a measured one. The Abstract and Conclusions carry the same statement.

Comment 3: Ramp limits are explicitly inactive for the original policy in the main 127-bus test, so the ramp term in the objective and the ramp projection in Equation 7 are never stressed, while the no-previous-action ablation exercises a constraint the main model never meets. The paper should add at least one scenario with steeper PV ramp, tighter R_i , or larger forecast error so the ramp constraint is active for the proposed policy too.

Response 3: We added Section 4.5 with a ramp-stress scenario on the same twelve 127-bus test days: R_i is tightened fourfold (0.045 to 0.01125 Mvar/h) with all injections, forecasts, and trained models unchanged; the ramp rate is not a node feature, so no retraining is involved. The ramp boundary becomes active for the proposed policy (1666 device-hours out of 12096; teacher 1860). Under stress, the four-head policy keeps zero voltage/current violations and zero screening failures, its daily loss stays 0.000429334 MWh above the teacher, and its objective gap (0.000646770) remains the smallest among the learned methods; the method ordering is unchanged. Two implementation adaptations are disclosed in Section 4.5: at sunrise/sunset the teacher's two-step quadratic program can become infeasible under the tightened ramp, so capability protection takes precedence over the inter-step coupling (mirroring the execution rule), and the QP solver is switched to an active-set method because the interior-point solver stalls on these instances. The scenario script (ramp_scenario.m), its outputs, and an independent verification script (qa_ramp.m, all checks passed) are added to the simulation package.

Dynamic Optimal Control of **Synthetic Radial** Distribution Networks Using Multi-Head Attention Graph Neural Networks

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Abstract. Hourly changes in distributed PV output and demand complicate Volt – VAR regulation on radial distribution networks. We construct synthetic 64-, 127-, and 253-bus networks that preserve total feeder length and nominal demand. Safely executed actions from a two-step rolling quadratic program supervise a two-layer, four-head graph-attention policy. All methods undergo the same constant-power AC screening and daily closed-loop evaluation. Across twelve 127-bus test days, the four-head policy has 1.610012582 MWh daily loss, 0.206110307 MWh below droop but 0.000374155 MWh above the teacher, with no violation. Its P95 full-step latency is 11.13526 ms, above the teacher's 7.67595 ms. **The loss gap is 0.023% of the teacher's daily loss, below utility relevance, while latency is 45% higher. Under a fourfold tighter ramp rate the ramp boundary becomes active for the proposed policy (1666 device-hours), and the loss gap stays 0.000429334 MWh with no violation. The results support synthetic radial Volt – VAR benchmarking only;** field networks and hardware latency remain unverified.

1. Introduction

Distributed photovoltaic output and stochastic demand make the injections and operating state of active distribution networks vary over time. Controllers must continually adjust resource outputs subject to voltage, line-capacity, and device constraints, while electrical coupling on a fixed radial topology makes actions at neighboring buses interdependent.

Graph policy networks have been applied to distribution-network Volt-VAR control, although their convergence need not be faster than that of vector-based policies [1]. Graph attention combined with deep reinforcement learning has enabled real-time multi-objective scheduling of active distribution networks [2]. Physics-assisted multi-agent graph learning **addresses topology changes [3], while hierarchical recurrent graph attention supports** decentralized Volt-VAR control under partial observations [4].

Further studies address reconfiguration coordinated with Volt-VAR control [5], robustness to observation perturbations [6], static optimal-power-flow approximation [7], voltage-sensitivity and topology priors [8], multi-timescale coordination [9], and multi-head active voltage control [10]. Combining these elements alone establishes no methodological distinction; the specific design must be evaluated under consistent physical constraints and scenarios.

This study addresses hourly inverter reactive-power control on fixed synthetic radial networks. The safely executed actions of a two-step rolling optimizer provide supervision for a four-head graph-attention policy, and every strategy uses the same device projection and nonlinear AC screening. Unlike work that jointly optimizes network reconfiguration [5] or coordinates multiple resources across time scales [9], we focus on one continuous control variable and examine closed-loop cost and online time. Relative to existing multi-head active-voltage control [10], we compare no-attention, single-head, and four-head policies under one screening rule and test changes in bus density within the same synthetic topology family. **We present the study strictly as a synthetic radial Volt–VAR benchmark and do not**

claim transfer to field large-scale networks, whose tap changers, capacitor switching, storage, unbalanced laterals, and measurement noise are outside the present model.

2. Network Model and Dynamic Control Problem

We construct a fixed synthetic radial network $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ with three feeders. Each feeder has L trunk buses, each with two single-bus laterals, giving $N = 1 + 9L$. Values $L = 7, 14, 28$ yield 64, 127, and 253 buses. The three networks retain the same total length and nominal demand per feeder; only bus density changes. There are no tie switches or topology control. The bases are 12.66 kV three-phase line voltage and $S_b = 1$ MVA; the root voltage is $V_{1,t} = 1 \angle 0$ pu. Each independent day has 24 hourly decisions. The demand shape is $g_t = 0.72 + 0.20e^{-(t-8)^2/12} + 0.36e^{-(t-19)^2/18}$ and the PV shape is $h_t = \max\{0, \sin[\pi(t-6)/12]\}$. Independent standard-normal daily load and solar factors are $1 + 0.08Z$ clipped to $[0.8, 1.2]$ and $0.85 + 0.12Z$ clipped to $[0.55, 1.05]$. Hourly load and PV multipliers have 3% and 6% normal perturbations clipped to $[0.9, 1.1]$ and $[0.85, 1.15]$, with PV output further capped at its peak. The next-hour forecast uses independent 10% noise clipped to $[0.7, 1.3]$ and does not use the future realized hourly disturbance. Complete days are kept separate across training, validation, and testing; all sequences are synthetic.

Let $P_{i,t}^L$ (MW), $Q_{i,t}^L = 0.35P_{i,t}^L$ (Mvar), and $P_{i,t}^{PV}$ (MW) denote active demand, reactive demand, and PV active output at bus i and time t . The inverter injection $q_{i,t}$ (Mvar) is zero at buses without an inverter, and each day begins with $q_{i,0} = 0$. On the energized tree directed from the root, constant-power AC flow uses complex bus voltage $V_{i,t}$, branch current $I_{ij,t}$, and per-unit branch impedance $r_{ij} + jx_{ij}$:

$$s_{i,t} = \frac{P_{i,t}^L - P_{i,t}^{PV} + j(Q_{i,t}^L - q_{i,t})}{S_b}, I_{ij,t} = \left(\frac{s_{j,t}}{V_{j,t}} \right)^* + \sum_{k \in \mathcal{C}(j)} I_{jk,t}, V_{j,t} = V_{i,t} - (r_{ij} + jx_{ij}) I_{ij,t}. \quad (1)$$

Here $\mathcal{C}(j)$ is the set of children of bus j , and the asterisk superscript in (1) denotes complex conjugation. Actual line loss is $P_{\text{loss},t} = S_b \sum_{(i,j) \in \mathcal{E}} r_{ij} |I_{ij,t}|^2$ MW. Inverter rating S_i^{inv} (MVA) determines its instantaneous reactive capability $q_{i,t}^{\text{max}}$ (Mvar). First clip the previous action to the present capability interval, defining $q_{i,t}^{\text{base}} = \text{clip}(q_{i,t-1}, -q_{i,t}^{\text{max}}, q_{i,t}^{\text{max}})$. The nominal ramp rate R_i (Mvar/h) and one-hour interval Δt constrain actions relative to this base. AC evaluation checks

$$q_{i,t}^{\text{max}} = \sqrt{(S_i^{\text{inv}})^2 - (P_{i,t}^{\text{PV}})^2}, |q_{i,t}| \leq q_{i,t}^{\text{max}}, |q_{i,t} - q_{i,t}^{\text{base}}| \leq R_i \Delta t, 0.95 \leq |V_{i,t}| \leq 1.05, |I_{ij,t}| \leq I_{ij}^{\text{max}}. \quad (2)$$

I_{ij}^{max} is a synthetic line apparent-power rating converted to per-unit current at 1 pu nominal voltage, rather than a field relay setting. For a line with n_l downstream load buses, the fixed rating is $S_l^{\text{max}} = \max\{0.15(14/L), 1.22\sqrt{1+0.35^2} n_l 0.06(14/L)\}$ MVA and $I_l^{\text{max}} = S_l^{\text{max}} / S_b$ pu. Inverters are assumed capable of supplying reactive power at night; auxiliary active-power loss is not modeled. If an increase in PV output makes the capability and ordinary ramp intervals disjoint, capability protection takes precedence and the forced derating is counted separately, as detailed in Section 3.

Table 1. Fixed parameters of the synthetic networks, control, and learning

Item	Setting
Size and bases	$L = 7, 14, 28$ gives $N = 64, 127, 253$; 12.66 kV, 1 MVA; three feeders
Lines	Total trunk length per feeder 7 km, each lateral $3.5 / L$ km; trunk $r / x = 0.35 / 0.40 \Omega / \text{km}$, lateral $0.45 / 0.35 \Omega / \text{km}$
Bus demand and devices	Each non-root bus has $P_i^0 = 0.06(14 / L)$ MW and $Q_i^0 = 0.35P_i^0$ Mvar; one inverter per trunk group, $P_i^{PV, \max} = 0.09(14 / L)$ MW, $S_i^{\text{inv}} = 0.11(14 / L)$ MVA
Control and limits	$R_i = 0.045(14 / L)$ Mvar/h; 24 one-hour intervals; voltage 0.95–1.05 pu; AC current rating derived from fixed synthetic capacity
Data and training	Sixty independent 127-bus days split 36/12/12 for training/validation/testing; eight independent test days each at 64 and 253 buses; two layers with four heads, eight hidden dimensions per head; batch size 32, learning rate 0.003, at most 25 epochs, early-stopping patience five epochs
Disturbances and seeds	Normal daily load/solar perturbation SDs 8%/12%, hourly load/PV 3%/6%, one-step forecast 10%; main-scenario/training/64-bus/253-bus seeds 42/1234/642/2532

3. Multi-Head Graph-Attentive Control Method

The teacher solves a two-step rolling quadratic program using current actual injections and a one-step forecast obtained only from known daily factors, the public next-hour shape, and independent forecast noise. Let $\mathbf{M}$ be the branch-by-downstream-bus matrix and $\mathbf{C}$ the inverter-to-bus mapping. The vectors $\mathbf{P}^L, \mathbf{P}^{PV}, \mathbf{Q}^L$ contain bus values in MW/Mvar, and $\mathbf{q}$ contains inverter actions in Mvar. Proxy branch active and reactive flows f_p, f_Q are per unit, and proxy voltages $\hat{V}$ are in pu:

$$f_p = \frac{\mathbf{M}(\mathbf{P}^L - \mathbf{P}^{PV})}{\mathbf{S}_b}, f_Q = \frac{\mathbf{M}(\mathbf{Q}^L - \mathbf{C}\mathbf{q})}{\mathbf{S}_b}, \hat{V} = \mathbf{1} - \mathbf{M}^{\text{topdiag}}(\mathbf{r})f_p - \mathbf{M}^{\text{topdiag}}(\mathbf{x})f_Q. \quad (3)$$

$\mathbf{r}$ and $\mathbf{x}$ contain per-unit line resistance and reactance. With $P_{\text{ref}} = 1$ MW, $V_{\text{ref}} = 0.05$ pu, and $Q_{\text{ref}} = 0.1$ Mvar, all terms in the objective are dimensionless. The first teacher action q_t^* is obtained from

$$\hat{P}_{\text{loss}, t} = \mathbf{S}_b \sum_l \mathbf{r}_l (f_{p,l,t}^2 + f_{Q,l,t}^2), J_t = \frac{\hat{P}_{\text{loss}, t}}{P_{\text{ref}}} + \frac{0.025}{N} \frac{\hat{V}_t - 1}{V_{\text{ref}}^2} + 0.0005 \frac{q_t - q_{t-1}}{Q_{\text{ref}}^2}, (q_t^*, q_{t+1}^*) = \arg \min_{q_t, q_{t+1}} (J_t + 0.8J_{t+1}) \quad (4)$$

The quadratic program constrains only the two-step device capability and action ramps, **with a one-step horizon** at the last hour **and** second-step capability from forecast PV. Equations (3)–(4) are a linear power-flow proxy, not an AC-feasible solution. Its first action passes the same AC safety screen as every comparator before it becomes an imitation label. The actual executed action updates the next control state. The daily aggregate objective in Table 2 retains the weights and scales of (4) **but uses** hourly AC loss and bus-voltage magnitudes, **summed** over 24 hours.

Node features include current demand and PV output, the noisy next-hour forecast, the previous action, current reactive capability, device type, trunk depth, and the AC voltage of the pre-action state after forced derating. Edge features are $\tilde{r}_{ij} = r_{ij} / 0.002$ and $\tilde{x}_{ij} = x_{ij} / 0.002$, with zeros on self-loops.

Let $z_i^{(0)}$ denote normalized node features and $\mathcal{N}(i)$ the physical neighbors. Each of the two graph-

attention layers has four heads of eight hidden dimensions and independently learned parameters. For head h , the score, neighborhood weight, and aggregation are

$$e_{ij}^{(h)} = \text{LeakyReLU} \left[\left(a_s^{(h)} \right)^{\text{top}} W^{(h)} z_i + \left(a_t^{(h)} \right)^{\text{top}} W^{(h)} z_j + \left(a_e^{(h)} \right)^{\text{top}} \left(\tilde{r}_{ij}, \tilde{x}_{ij} \right) \right], \alpha_j^{(h)} = \frac{\exp_{ij}^{(h)}}{\sum_{k \in N(i) \cup \{i\}} \exp_{ik}^{(h)}}, z'_i = \text{concat}_{h=1}^4 t_i \quad (5)$$

The operator `concat` joins the head outputs, and `LeakyReLU` has negative slope 0.2. A shared readout with `tanh` produces normalized reactive actions $\hat{a}_t$. The safety-screened teacher actions are divided by their instantaneous $q_{i,t}^{\max}$ to form labels a_t^* . Training minimizes the squared error over training times $\mathcal{T}$ and inverter buses $\mathcal{D}$:

$$J_{\text{fit}}(\theta) = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left(\hat{a}_{i,t}(\theta) - a_{i,t}^* \right)^2. \quad (6)$$

θ contains all trainable parameters. At inference, $\hat{q}_{i,t} = \hat{a}_{i,t} q_{i,t}^{\max}$ converts network output to Mvar. The previous action is first clipped to the present capability interval, yielding the ramp reference $q_{i,t}^{\text{base}}$. The network proposal $\hat{q}_t$ is then projected onto the device intersection $\mathcal{Q}_t$. With projection Π and $\Lambda = \{1, 0.75, 0.5, 0.25, 0\}$, the safety-screen candidates are

$$q_{i,t}^{\text{base}} = \text{clip}(q_{i,t-1}, -q_{i,t}^{\max}, q_{i,t}^{\max}), \mathcal{Q}_t = \prod_{i \in \mathcal{D}} \left(\left[-q_{i,t}^{\max}, q_{i,t}^{\max} \right] \cap \left[q_{i,t}^{\text{base}} - R_i \Delta t, q_{i,t}^{\text{base}} + R_i \Delta t \right] \right), q^{(\lambda)} = \lambda \Pi_{\mathcal{Q}_t}(\hat{q}) + (1 - \lambda) \Pi_{\mathcal{Q}_t}(q_{i,t}^{\text{base}}) \quad (7)$$

For λ in descending order, each candidate in (7) is evaluated by AC flow **under** the current realized injections; **the first candidate** satisfying (2) is executed, **else** the least-violating **one, with** failure recorded. Screening **guarantees no** feasibility and **applies identically to every comparator**; **both** attention models **keep** the total hidden width of 32. **only** continuous inverter reactive output is controlled.

4. Simulation Experiments and Results

4.1 Experimental Setup and Metrics

Independent days are generated using Table 1; only days 49–60 form the main test set. The four-head model is trained on the teacher's executed trajectories. Comparators are the two-step teacher, local Volt–VAR droop, 32-dimensional graph convolution without attention, single-head attention, and an ablation without the previous-action feature. Every method **uses the safety screen in (7) under** the same test injections and causal forecast. We report daily AC loss (MWh), the normalized objective evaluated on actual AC states, bus-voltage and line-current violation hours, safety-backtracking/failure hours, and online latency including feature construction, screening, and AC solves; teacher latency additionally includes the quadratic-program solve. Results are compared by paired test day with 95% intervals. The 127-bus case has 12 test days; the 64- and 253-bus cases use independent eight-day **pools** with different **seeds**.

4.2 Main Test and Physical Constraints

Table 2 summarizes the twelve 127-bus test days. The four-head model has a daily loss of 1.610012582 MWh, which is 0.206110307 MWh below local droop; the 95% paired t interval for its difference is $[-0.209023340, -0.203197273]$ MWh. Its loss is 0.000374155 MWh above the teacher, with interval $[0.000319620, 0.000428689]$ MWh. Its daily aggregate objective exceeds the teacher by

0.000504357, with interval $[0.000303832, 0.000704882]$. Neither the four-head model nor the teacher has a voltage/current violation or screening failure over the 288 test hours. Droop has three line-overload hours and three screening failures. Figure 1 shows day 55, near the median teacher daily loss. Figure 2 shows these paired differences in full.

Table 2. Closed-loop results for twelve 127-bus test days (violation/failure totals over 288 h)

Method	Daily loss/(MWh)	Daily aggregate objective	Voltage/current violation hours	Screening failures/h	P95 latency/ms
Two-step teacher	1.609638427	1.646692015	0/0	0	7.67595
Local droop	1.816122889	1.884218620	0/3	3	1.07771
Four-head graph attention	1.610012582	1.647196372	0/0	0	11.13526
Graph convolution without attention	1.610865040	1.647549964	0/0	0	2.04780
Single-head graph attention	1.610548652	1.647651546	0/0	0	3.69514
Four-head, no previous-action feature	1.609256800	1.647817028	0/0	0	9.98725

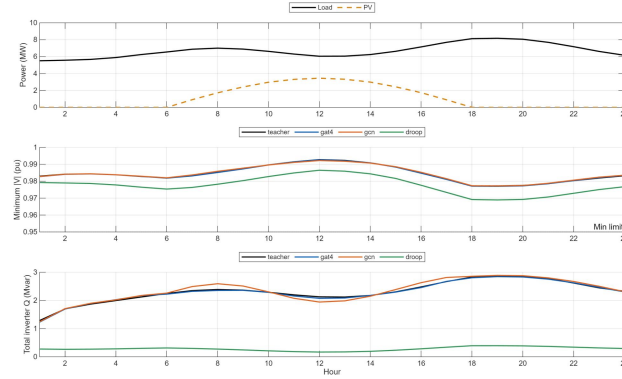


Fig. 1. Power profiles, minimum bus voltage, and total reactive injection on a representative 127-bus test day.

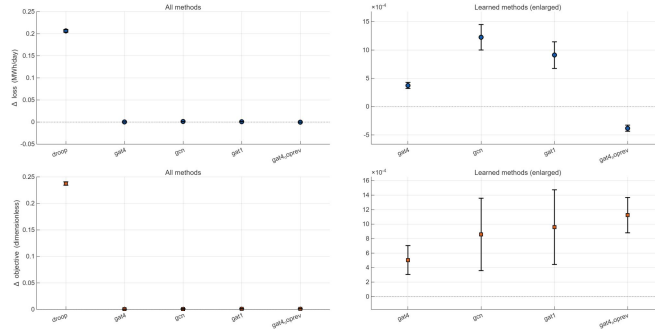


Fig. 2. Paired mean differences relative to the teacher over twelve 127-bus test days with 95% t intervals; rows show loss (MWh/day) and the daily aggregate objective; columns show the full range and an enlarged learned-method view.

4.3 Attention, State Features, and Time Cost

The four-head model loses 0.000852458 MWh/day less than graph convolution without attention, with paired interval $[-0.001035782, -0.000669134]$ MWh, and 0.000536070 MWh/day less than single-head attention, with interval $[-0.000726719, -0.000345421]$ MWh. Removing the previous-action feature yields 0.000381627 MWh/day less loss than the teacher, yet its daily aggregate objective is 0.001125013 higher and the ramp boundary is active in 170 device-hours, rather than violated 170 times. Loss alone therefore misses the control-movement cost. Its 127-bus P95 full-step latency is 11.13526

ms, above the teacher's 7.67595 ms. The ramp boundary is inactive for the original model in the main test; Section 4.5 stresses it directly.

The gap to the teacher is 0.023% of its daily loss and the advantage over graph convolution is 0.053%, both below utility relevance, while the four-head policy pays 45% higher P95 latency than the teacher and 5.4 times graph convolution's. We therefore do not present head-count scaling as a finding: the remaining differences are operationally negligible, and whether they justify the extra inference time is a deployment question, not a measured one.

4.4 Same-Family Size Behavior and Limits

Within each size, the four-head model's daily loss differences from the corresponding teacher are -0.000620153 , 0.000374155 , and 0.001022922 MWh, respectively. Latency is lower than the teacher at 253 buses but higher at 127 buses; however, **graph convolution has a lower 253-bus objective difference**. At 64 buses, the four-head objective difference of 0.000148478 has a 95% interval $[-0.000297824, 0.000594780]$ spanning zero. **Cross-size** loss differences cannot be attributed to bus count **given the different day pools**. The four-head model has no recorded screening failure at any **size** within the present synthetic scenarios and forecast noise; **nighttime** auxiliary loss, field communication, and field operation **are unverified**.

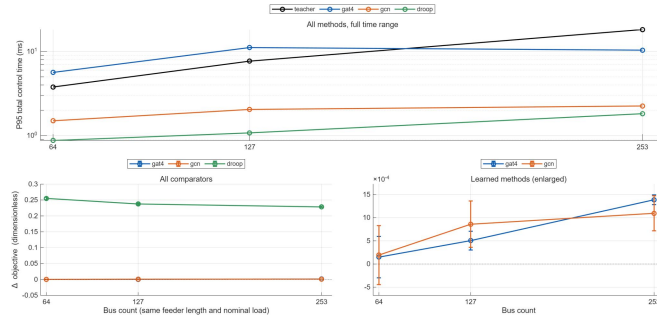


Fig. 3. P95 full-step latency (top), the full range of daily objective differences relative to the teacher within each size (lower left), and an enlarged learned-model view (lower right) with 95% intervals; 64/253-bus cases use independent eight-day pools and the 127-bus case a twelve-day pool, so cross-size results are not paired.

4.5 Ramp-Stress Scenario

A supplementary scenario stresses the ramp constraint on the same twelve 127-bus test days: the ramp rate is tightened fourfold, from 0.045 to 0.01125 Mvar/h, with injections, forecasts, and trained models unchanged; the ramp rate is not a node feature, so no retraining is involved. At sunrise and sunset the teacher's two-step quadratic program can become infeasible under the tightened ramp; as at execution time, capability protection then takes precedence over the inter-step coupling, and the solver is switched to an active-set method.

The tightened ramp activates the boundary for every optimization-based policy: 1860 device-hours for the teacher and 1666 for the four-head policy out of 12096, against zero in the main test. The four-head policy keeps zero voltage/current violations and screening failures; its daily loss is 0.000429334 MWh above the teacher (95% paired interval $[0.000377182, 0.000481485]$ MWh), and its daily aggregate objective exceeds the teacher by 0.000646770 (interval $[0.000453550, 0.000839991]$), the smallest objective gap among the learned methods as in the main test. Graph convolution and single-head attention remain further from the teacher (loss gaps 0.001223530 and 0.001014616 MWh), droop is unaffected because its actions rarely reach the bound, and the no-previous-action ablation again trades a slightly lower loss (0.000368116 MWh above the teacher)

for a higher objective (0.000814629) and more ramp-active hours (1842). The tightened ramp therefore changes neither the method ordering nor the conclusion that the loss differences are small.

5. Conclusions

On the synthetic 127-bus radial test, the four-head policy has no violation after common AC screening. Its daily loss approaches the teacher and improves on local droop and two graph-model comparators with differences below utility relevance; its 127-bus P95 latency is 45% higher than the teacher's, so head count is not claimed as an advantage. Latency is lower than the teacher only at 253 buses, where graph convolution has a lower objective gap; the 64-bus objective interval spans zero. The no-previous-action ablation shows that loss alone hides control-movement cost. A fourfold tighter ramp activates the boundary for the proposed policy (1666 device-hours) without changing the method ordering or adding violations.

Acknowledgements

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